By making the interferometer's arms a thousand times longer than the bar (a few kilometers rather than a few meters), one can make the waves' tidal forces a thousand times bigger and thus improve the sensitivity of the instrument a thousand-fold.[8] The bar, by contrast, cannot be lengthened much. A kilometer-Iong bar would have a natural frequency less than one cycle per second and thus would not operate at the frequencies where we think the most interesting sources lie. Moreover, at such a low frequency, one must launch the bar into space to isolate it from vibrations of the ground and from the fluctuating gravity of the Earth's atmosphere. Putting such a bar in space would be ridiculously expensIve.

Because it is a thousand times longer than the bar, the interferometer is a thousand times more immune to the "kick" produced by the measurement process. This immunity means that the interferometer does *not* need to circumvent the kick with the aid of a (difficult to construct) quantum nondemolition sensor. The bar, by contrast, can detect the expected waves only if it employs quantum nondemolition.

If the interferometer has such great advantages over the bar (far larger bandwidth and far larger potential sensitivity), then why didn't Braginsky, Weber, and others build interferometers instead of bars? When I asked Braginsky in the mid-1970s, he replied that bar detectors are simple, while interferometers are horrendously complex. A small, intimate team like his in Moscow had a reasonable chance of making bar detectors work well enough to discover gravitational waves. However, to construct, debug, and operate interferometric detectors success fully would require a huge team and large amounts of money—and Braginsky doubted whether, even with such a team and such money, so complex a detector could succeed.

Ten years later, as the painful evidence mounted that bars would have great difficulty reaching 10^{-21} sensitivity, Braginsky visited Caltech and was impressed with the progress that Drever's team had achieved with interferometers. Interferometers, he concluded, will probably succeed after all. But the huge team and large money required for success were not to his taste; so upon returning to Moscow, he redirected most of his own team's efforts away from gravitational-wave detection. (Elsewhere in the world bars have continued to be developed, which is fortunate; they are cheap compared to interferometers, for now they are more sensitive, and in the long run they might play special roles at high gravity-wave frequencies.)